

BrickAtlas Display-Grounded Evidence Following Scientific Supplement

Anonymous CVPR submission

Paper ID Draft

1. Reading Guide and Version Boundary

This supplement documents the display-v3 observation contract, human qualification, response statistics and graph audit. The full 617 historical question dossiers are distributed separately as `dossiers.json` and a searchable HTML interface. They are not 617 new display-v3 questions. Historical prompts and answers remain historical even when a parent ID is linked to new observations.

The analysis version is `ldraw2-evidence-v3`; the visual dataset is `brickatlas-display-v3`. Color's 146 endpoints, the 67 panels, graph inputs and historical dossiers retain their earlier bytes. A separate parent-mapping overlay supplies current pair IDs; old dossier links remain in the original JSON. Both primary Part-type arms are rebuilt. Canonical source identity is the source hash and placed-instance ID. Display references resolve within one current image. The complete four-part identity rule and all parent-to-observation mappings are released with the dataset. No old model answer or review is transferred to a new prompt.

The artifact has three distinct inspection modes. The evaluator site exposes pairing, gold and provenance for scientific audit. The blind review packages expose only one exact observation at a time, without gold or pair membership. The historical 3D viewer allows rotation, zoom, source-part isolation, expansion and visibility replay. Only the blind packages implement the qualification protocol; richer 3D access cannot substitute for judging native model-input pixels.

2. Exact Visual Stimuli

2.1. Reference questions and a concrete pair

For the OMR 42004 Color example, both arms ask:

What color is the object currently marked B0035 in this image?

Both include the rule that references select current image objects and that appearances or labels may have been

edited. Gold changes from Blue (choice D) to Red (choice C). Original source truth stores Black. This is an explicit rendered-appearance counterfactual, not a claim that the source record changed.

For the corresponding Part-type example, both arms ask:

Which displayed candidate label currently marks an object with the same visible part geometry as the target marked B0035? Ignore color and pose. Answer from this image, not from historical source identities.

Candidate B0036 in A has the same source type, 32250, as the target. After the B0036/B0116 candidate-reference swap, B0116 selects that same matching geometry in B. The option-ID gold changes from C to D. This describes generated golds only. There is no observed model C-to-D or C-to-C response case in the current release.

Figure 1 in the main paper uses the actual native PNGs with an identical crop rectangle for the two arms within each family. Its annotations are paper-only. `figure-provenance.json` records the four observation IDs, full PNG hashes and crop rectangles. Full images and full prompts remain available through the paired-observation inspector.

2.2. Rendering audit

All primary and auxiliary images have a native resolution of 1280×800 . Color edits keep the geometry, pose, camera, label projections, visibility and material settings identical. The body mask is the projected part body; one pixel of dilation accommodates antialiasing at the silhouette. Changed pixels outside that mask number zero in every Color pair. The mask test is not a measurement of human color separability, projected difficulty or assembly stability.

The 73 pairs use two distinct named opaque colors each; both arms may differ from the historical source appearance. Aggregate category counts are 37 Red, 37 Blue, 36 Yellow and 36 Black. Canonical color, part type and transforms are preserved separately. The donor feasibility inventory retains every candidate considered for the older mechanism

and explains why a general donor swap cannot be relabeled as a color-only intervention.

Part-type regenerates all 134 primary images after joint option/swap assignment. Within every pair the underlying geometry render is byte-identical, with pixel changes confined to label masks. Two independent regeneration canaries are byte-identical. Source poses, camera and anchor projections remain fixed. Independent gold validation reconstructs the display mapping and source part-type equality; visual uniqueness remains a human judgment.

2.3. Position-reference control

The 67 auxiliary observations each have a target panel and four candidate panels, without labels inside the image (Figure 1). The question describes which panel is the target and the ordering of the four candidates. The correct option is derived independently from source geometry/type correspondence. No candidate-reference change pair is manufactured from these panels.

Each auxiliary observation requires two independent raw human reviews, for 134 additional judgments. Its final decidable set and accuracy are separate from the 280 primary endpoints and their paired Both. Comparing it with a label-based arm changes several factors at once: reference interface, separation of objects and view access. A higher score would not isolate geometry skill, and a lower score would not isolate position understanding.

3. Finite Assignment and Answer Priors

The candidate inventory records visible operand IDs, source types, source bounding-box centers and projected label anchors. All 24 permutations of four options are paired with each admissible nonmatching swap, preserving unique source-type matching. The 4,824 configurations are ordered by parent, option-label tuple and swap instance. Constraint-equivalent configurations retain the smallest integer hash cost, leaving 2,412 binary variables.

The frozen constraints require exactly one configuration per parent, 16–17 golds per arm/position, 5–6 per off-diagonal transition, fixed-choice source accuracy at most .30, and each declared oracle-A rule’s micro/source accuracy at most .40. SciPy 1.17.1/HiGHS, one thread and zero relative gap return optimum 1,007,668. The checked assignment file is authoritative; its contract, inventory and code hashes remain available. There is no model-result-based seed search.

Table 1. Transition counts: rows are A gold, columns B gold. Left: frozen Color (73). Right: optimized Part-type (67).

A\B	A	B	C	D	A\B	A	B	C	D
A	0	3	3	7	A	0	6	5	5
B	7	0	7	8	B	6	0	6	5
C	4	8	0	7	C	5	6	0	6
D	5	4	10	0	D	6	5	6	0

Table 2. All 13 declared Part-type oracle-A attacks, B accuracy (%). A is supplied correctly, so NewAcc=Both. Each rule excludes the A-matching candidate and never reads B. Source intervals are descriptive after design optimization, not out-of-sample guarantees.

Oracle-A rule	Micro	Source	Source 95% CI
First alternative	34.33	26.94	[15.62, 38.61]
Last alternative	32.84	32.01	[19.93, 44.86]
Cyclic next	35.82	38.68	[25.00, 53.33]
Cyclic previous	34.33	28.89	[17.92, 40.35]
Cyclic two	29.85	32.43	[19.79, 46.25]
Smallest alternate label	29.85	24.58	[14.58, 34.38]
Largest alternate label	38.81	39.03	[26.04, 52.50]
Nearest label number	29.85	30.42	[18.75, 43.12]
Farthest label number	31.34	29.03	[17.08, 41.53]
Nearest source center	29.85	36.60	[23.61, 50.97]
Farthest source center	35.82	28.75	[17.92, 40.21]
Nearest projected anchor	25.37	30.14	[17.36, 44.38]
Farthest projected anchor	32.84	27.08	[16.25, 38.75]

Option rules select first/last alternatives or cyclic positions. Label rules select smallest/largest alternate numbers or closest/farthest numbers from the target. Spatial rules select nearest/farthest source centers or projected anchors. Ties use ascending original numeric labels. Spatial audits additionally receive A-side source geometry: they are privileged construction attacks, not image-only baselines. The independent auditor reconstructs candidates/golds from source records and rendering metadata, not the optimizer’s hit counts.

Both families also report fixed-choice and smallest/largest option-label text rules with A accuracy, NewAcc, Both, every prediction/error, source intervals and author sensitivity. The old first-alternative attack is archived at 67/67, versus 23/67 now. Color was not optimized; its largest declared oracle-A micro score is 32/73. Neither family claims universal shortcut immunity.

4. Human Review Protocol

4.1. Sampling universe and blinding

Six initial packages contain 115, 116, 114, 115, 115 and 119 units. Their union covers 146 Color endpoints, 134 Part-

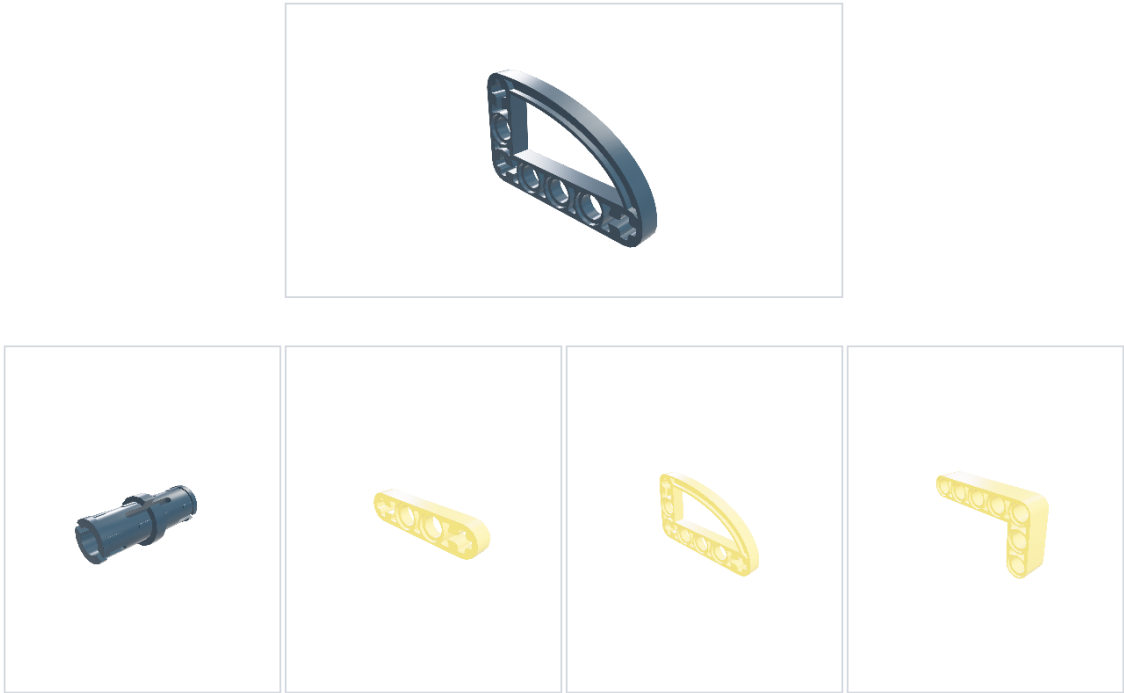


Figure 1. Complete position-reference control image for the OMR 42004 example. The upper panel is the target and the four lower panels are candidates referenced by ordinal position in the question. No text is embedded in the image. The correct candidate is third from the left among the four candidates. These isolated source parts are not a proposed assembly. This interface changes layout and visible access, so its contrast with candidate labels is descriptive, not an identified OCR effect.

type endpoints and 67 position controls exactly twice. Each reviewer sees at most one observation per parent, including panels. Three-observation parents use all six people without exposing counterparts. Assignments cross arm/slot combinations and hide condition names. The validator checks displayed text/options and image bytes against each frozen wire.

Reviewers work independently before model inference and do not receive pair membership, condition labels, canonical source records, gold, model responses or another person’s answers. The coordinator distributes one package to each of six different people. The native image is scrollable rather than silently downsampled. A reviewer attests that the full native observation was inspected.

4.2. Raw and final records

- Every raw record contains:
- opaque unit ID and exact wire-observation hash;
 - reviewer pseudonym, timestamp and independence attestation;
 - decidable, ambiguous, unanswerable or contradictory decision;
 - chosen visual option where applicable;
 - four Boolean reference/answer/visibility/contradiction checks;

- reason code, free-text reason and native-size attestation.
- A decidable record requires a listed option, all four checks passing, and the no-exclusion reason code. Nondecidable records require a reason. Raw records are never overwritten by a final decision. Duplicate raw reviews, reuse of a reviewer identity across the six slots, obsolete hashes and synthetic evidence are rejected.
- If categorical decisions, choices and checks agree, the record is finalized under the frozen rule. Different explanatory prose alone does not constitute a categorical disagreement. Any decision, choice or check disagreement requires an additional person’s adjudication. The adjudicator sees the single observation and the two judgments, but not gold, the other endpoint or model responses. A reason is required. Up to three extra adjudicator slots receive disjoint parents; they cannot be initial reviewers or see any other view of a parent. Final import reconstructs decisions from retained original exports. A final decidable choice conflicting with the generated gold returns the item to pending and requires dataset revision.

4.3. Agreement and gate semantics

For doubly reviewed units, raw decision agreement is the fraction of equal categorical decisions. Descriptive pooled

Table 3. All source assemblies. Band is source scale. Color and Part-type count primary parents, each contributing exactly one A/B pair. All 24 sources appear in both visual families; variants do not increase source count.

Source	Name	Band	Parts	Color	Part-type
31028	Sea Plane	D1	53	2	2
31027	Blue Racer	D1	67	2	2
10156	LEGO Truck	D1	111	2	1
omr-42102	Mini CLAAS XERION	D1	129	2	1
omr-42020	Twin-Rotor Helicopter	D1	141	2	1
10036	Pizza To Go	D2	166	2	2
10014	Caboose	D2	170	2	2
omr-31031	Rainforest Animals	D2	215	2	2
omr-31088	Deep Sea Creatures	D2	230	2	1
omr-42004	Mini Backhoe Loader	D2	246	2	2
omr-42061	Telehandler	D2	261	2	2
31009	Small Cottage	D2	270	2	1
5867	Super Speedster	D2	278	2	2
10128	Train Level Crossing	D2	336	2	2
10159	City Airport	D3	592	3	3
10001	Metroliner	D3	848	4	4
omr-10269	Harley-Davidson Fat Boy	D4	1029	4	4
omr-10242	MINI Cooper	D4	1086	4	4
omr-42066	Air Race Jet	D4	1244	5	5
10213	Shuttle Adventure	D4	1254	5	5
omr-10265	Ford Mustang	D4	1357	5	5
10220	Volkswagen T1 Camper Van	D4	1428	5	4
omr-21309	NASA Apollo Saturn V	D4	1845	5	5
omr-42054	CLAAS XERION 5000 TRAC VC	D4	1978	5	5

two-slot kappa is

$$\kappa = (p_o - p_e)/(1 - p_e), \quad (1)$$

where p_e is computed from the two empirical decision marginals. Assigned people vary across units; this is not the reliability coefficient of one fixed pair of reviewers. No doubly reviewed units yields null; degenerate marginals with $p_e = 1$ also yield null. Choice agreement and each structured check have separate statistics. Primary and auxiliary counts are explicit. Adjudicated agreement does not replace raw agreement.

Before real inference, every planned endpoint needs a final nonpending decision. This gate does not require every endpoint to be decidable: ambiguous, unanswerable and contradictory outcomes remain part of the full benchmark report. Additional QA-subset estimates require both endpoints to be finally decidable. Missing either A or B blocks that pair’s inclusion. The final summary is reconstructed from retained raw inputs; editing a summary cannot satisfy the gate.

At release, raw human reviews are 0/560 for primary endpoints and 0/134 for auxiliary controls. Final primary decisions are 0/280; both-decidable pairs are 0/140. These are unresolved acceptance requirements. Software tests exercise the rule with temporary synthetic fixtures; none is im-

ported as a real review.

5. Statistics and Genuine Response Evidence

5.1. Outcomes and denominators

For a changed-answer pair, B outcomes are mutually exclusive: correct new answer, valid old answer, another valid answer, or invalid. Their micro proportions sum to one over all planned pairs. A accuracy and Both additionally distinguish the A endpoint. Missing calls do not remove the pair. JSON validity is reported for each arm and for both, not inferred from semantic correctness alone.

Conditional adaptation uses only A-correct pairs. Micro adaptation is the both-correct count divided by that count. Source-macro adaptation first computes each source’s conditional rate, then averages over sources with at least one A-correct pair. The report includes both eligible pair and eligible source counts. Those two estimands can differ; neither replaces unconditional Both.

For invariance graph pairs, $y_A = y_B$. Both remains the joint success criterion. A repeated wrong answer is consistent but incorrect. Stability of raw invalid strings has no semantic meaning. The current visual release does not import older camera/ID-renaming invariance responses under the new display contract.

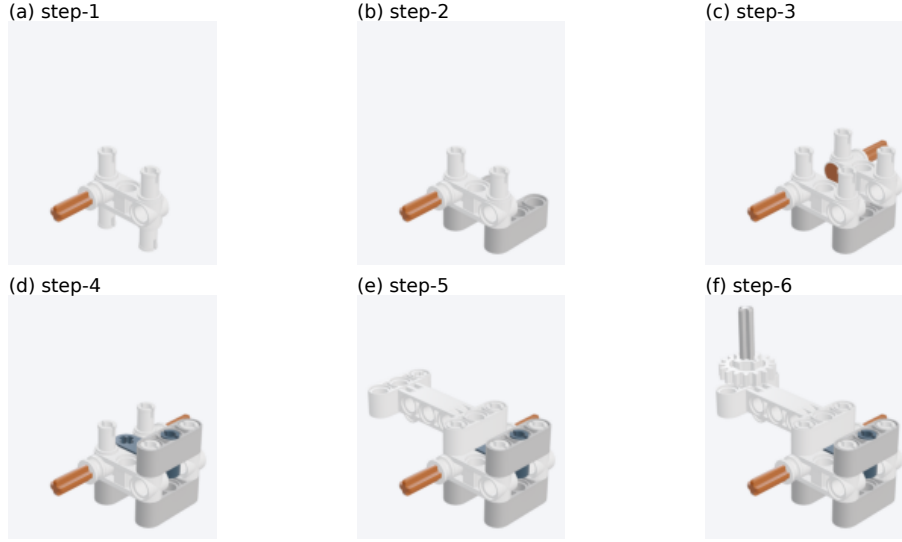


Figure 2. Historical given-order visibility replay for set 42102, retained as a 3D inspection example. Six accepted actions reveal parts at original source transforms. The sequence is supplied by the control, not discovered by a model. Display correspondence does not demonstrate a collision-free insertion path or gravity stability. The original figure hash is retained in publication provenance.

5.2. Sources, subset shifts and rank comparisons

Source-macro estimates give equal weight to sources after averaging their within-family pairs. Whole-source resampling uses 10,000 draws and a fixed seed, 20260920. Paired model differences use the same items and resampling weights. The QA-minus-full calculation recomputes both means using shared source draws and the subset’s actual source coverage. A draw containing no eligible subset source is recorded as undefined; it is not assigned a zero score.

Rank points are computed using exact rational source means. Equal values receive midranks, avoiding arbitrary floating-point thresholds for ties. The rank comparison reports A, NewAcc and Both, the sign of each paired difference, point reversals versus ties, descriptive intervals and agreement of directions across full/QA sets. With many families or models, these unadjusted comparisons remain exploratory. An interval containing zero supports neither directional ranking nor equivalence. The estimand is a single complete invocation set; no interval here measures cross-call provider variability.

5.3. Author and contributor sensitivity

Sources group by their first credited model author, removing bracketed handles and retaining OMR editors separately: 18 primary authors and 17 contributor-connected clusters. The latter connect any sources sharing an author/editor; source 10014 joins Robert Paciorek’s cluster via OMR editing. Table 4 lists the repeated primary authors.

Table 4. Repeated primary authors: 10/24 sources in four groups. Other primary-author groups each contain one source.

Repeated primary author	Source IDs
Merlijn Wissink	31027, 31028
Philippe Hurbain	42054, 42061, 42066
Robert Paciorek	10036, 10128, 10156
Zoltan Keri	10001, 10159

For within-source mean m_s and eligible sources S_a of author a , the sensitivity mean is $|A|^{-1} \sum_a |S_a|^{-1} \sum_{s \in S_a} m_s$. Ten thousand whole-author draws (seed 20260920) recompute equal-author and equal-source means with shared draws, including their difference. Each author’s omission yields source/author means; paired model differences flag changed directions. The procedure repeats for contributor clusters and reports actual eligible source/group counts for each subset.

All 276 source pairs have part-number count-multiset and set Jaccards. Table 5 shows the five largest count overlaps. No spatial alignment is performed; similar inventories need not share geometry.

Table 5. Largest inventory overlaps. Count J. is $\sum_t \min(n_{at}, n_{bt}) / \sum_t \max(n_{at}, n_{bt})$; Set J. is intersection/union of distinct part types.

Source A	Source B	Count J.	Set J.
42054	42066	0.450	0.511
42004	42061	0.370	0.354
42004	42020	0.325	0.431
10220	10242	0.322	0.257
10265	10269	0.280	0.320

5.4. Response cases and authenticity boundary

Only the exact roster of nine terminal live runs can enter empirical analysis, including explicitly closed failures. All visual/no-image runs share one finalized QA snapshot. Run IDs, adapter files/hashes, inclusion reasons and requested/expected revisions are frozen. Missing or extra runs block analysis; changing a model requires a new study version. The runner records raw response bytes, extracted output text, normalized choice, model revision and full-request hash. Analysis reparses raw responses; it does not trust manually entered accuracy columns. Missing or malformed responses remain failures. Requests and intents precede network calls. Interruption closure distinguishes unattempted requests from attempted calls without receipts; both remain in the 2,739 scheduled denominator. Closure records are not provider responses. The model-run index binds each roster entry, requests, receipts, QA, run dates and report rows. Current completed/planned runs are 0/9.

The no-image control removes only image content from all 347 visual messages per model. It retains question/options and measures evidence dependency. Its family-specific outcomes and paired visual-minus-no-image differences remain separate from graph scores.

Case extraction uses genuine QA-decidable pairs, separately by family and model. It looks for A-correct to new gold, A-correct to old gold, and the same valid wrong answer at both endpoints. Selection is deterministic by hash order within a case type. An absent type stays absent, and its absence is not filled with a constructed response. At release there are no eligible real cases and no empirical rank comparison. The example in Section 2 describes stimulus semantics only.

The collection pipeline checks hashes, provenance and declared response models. Local files and attestations alone cannot cryptographically prove that a remote provider generated a receipt or that two humans worked independently. Retained provider records and coordinator oversight complement the machine checks.

6. Graph Partitions and Algorithmic Results

6.1. Full strong-pair inventory

Table 6 lists all 12 strong changed-answer pairs. Every row matches surviving nodes, edge count, target degree, complete degree multiset and isolate count. BFS and union-find are both correct on every row. Anonymous degree and 1-WL feature predictors are jointly incorrect on every row because their restricted features do not distinguish the two golds.

Table 6. Every strong change pair; 12 pairs from eight sources. The final column is the exact per-pair result, not an average over generated variants. BFS and union-find have Both=1 on every row. The machine audit additionally supplies full feature signatures, source-held-out lookup support and predictions.

Parent task	y_A	y_B	Isolates+1 Both
ld2-10001-graph-removal-4	2	3	0
ld2-10014-graph-removal-1	2	3	0
ld2-10014-graph-removal-2	1	2	0
ld2-10213-graph-removal-1	4	3	0
ld2-31009-graph-removal-2	4	3	0
ld2-5867-graph-removal-1	1	2	0
ld2-omr-10269-graph-removal-1	2	1	0
ld2-omr-10269-graph-removal-2	2	1	0
ld2-omr-10269-graph-removal-4	2	1	0
ld2-omr-21309-graph-removal-2	2	3	0
ld2-omr-21309-graph-removal-4	2	1	0
ld2-omr-42054-graph-removal-4	3	4	0

Table 7. Natural graph source-macro accuracy in percent, with descriptive whole-source percentile intervals. All 73 items and 24 sources are retained. Lookup training excludes the test source and is separate from hypothetical graphs.

Algorithm	Source macro	95% interval
Node-count lookup	83.75	[73.12, 92.92]
Edge-count lookup	61.11	[46.39, 75.21]
Degree-histogram lookup	93.96	[87.71, 99.17]
Anonymous 1-WL lookup	93.96	[87.71, 99.17]
Isolates + one	97.92	[93.75, 100.00]
BFS	100.00	[100.00, 100.00]
Union-find	100.00	[100.00, 100.00]

6.2. Natural distribution and invariance

Table 7 supplies natural source intervals; the main paper reports exact item counts. A node-count *lookup* is distinct from the older hard-coded $|V| - 1$ predictor: it learns a feature-to-answer table on other sources. The distinction is preserved in filenames, labels and reports.

Table 8 gives both invariance partitions. The 73 anchors are reused from the change pairs, so counting all

pair endpoints would yield 292 appearances but only 219 distinct observations. The lookup training population includes changed endpoints once and excludes isomorphic invariance copies. This prevents those copies from disproportionately weighting a source label.

Table 8. Both-correct on invariance pairs. Strong and degree-visible sources overlap with the corresponding changed-answer partitions. Stable predictions need not be correct.

Algorithm	Strong / 12	Degree-visible / 61
Node-count lookup	6	31
Edge-count lookup	7	32
Degree-histogram lookup	6	61
Anonymous 1-WL lookup	6	61
Isolates + one	6	61
BFS	12	61
Union-find	12	61

6.3. Answer balance and small-sample limits

The strong direction count is six increases and six decreases. Table 9 discloses that raw arm answer frequencies are not identical. Both is invariant to swapping endpoint roles. Symmetric endpoint accuracy averages the two endpoints within each existing pair. Optional equal arm weighting uses each endpoint in both arm roles with half weight; it changes neither observations nor pair/source counts. It does not create uniform label frequencies.

Table 9. Stored strong-change answer histograms. Equal numbers of increases/decreases do not imply equal per-answer arm frequencies.

Stored arm	Gold 1	Gold 2	Gold 3	Gold 4
A	2	7	1	2
B	4	2	5	1

The Wilson intervals $[0, .2425]$ and $[.7575, 1]$ for 0/12 and 12/12 are approximate because source reuse violates pair independence. Boundary whole-source bootstraps may collapse to a point. The released intervals and per-pair data expose these limitations rather than imply a precisely known population effect.

All strong graphs instantiate a narrow construction: six-cycle versus two triangles, plus identical isolates. Uniform-feature anonymous 1-WL cannot distinguish them at any refinement depth. A degree or 1-WL lookup cannot evade this collision by fitting labels more aggressively. A model with unique-ID features, positions, spectral information or higher-order computation is outside this feature restriction. No trained GNN or language-model result is claimed by the algorithmic audit.

7. Source Coverage and Historical Controls

Table 3 makes source coverage explicit. Source files come with recorded URLs, author attribution, license metadata and content hashes in the historical catalog. Dataset-level paired analysis cannot substitute for source-level split discipline. Any future training/evaluation split must keep each source and all its derived views, interventions and controls together. Common part libraries may still create dependencies across such splits.

The historical families remain an inspection resource. Distance reads coordinates and computes a value; Interface and Coverage read supplied records; Step lookup reads author annotations. Neighbors and Graph deletion operate the explicitly supplied graph. Evidence limit and Step sequence include fixed-answer controls. Restoration follows a visibility contract at a saved source pose. None establishes physical insertion planning.

The viewer can replay these actions, expose the original graph/records and inspect part IDs. Expansion is an inspection transformation, not a physically valid final arrangement. Isolation omits supports without changing the saved assembly. Source-local overlap candidates still require physical review. The new color and label interventions do not claim to have corrected the historical source geometry.

8. Reproduction and Acceptance Checklist

The source package contains this manuscript, its main paper, generated tables, figures, bibliography and the CVPR style. Both PDFs compile with Tectonic. The repository additionally contains the publication generator, frozen stimuli, review packages, validators and raw-input hashes. The build guide specifies the checked runtime and commands.

The following machine checks are complete in this release: 347 frozen native visual observations; 140 paired gold changes; 73 controlled body-pixel differences; all visual golds independently recomputed; 219 graph request snapshots and dual-algorithm gold checks; all 2,739 planned request envelopes; and 27 new protocol/analysis tests. The 26 historical tests remain preserved. Browser and PDF inspection validate delivery behavior and layout. Those inspections are not human visual-task certification.

Empirical acceptance remains open for the 560 primary human reviews, 134 auxiliary reviews and any necessary adjudications, followed by new model inference. The required analysis must show full and QA-decidable results, valid-format rates, family-specific response transitions, paired source intervals and genuine case examples where they exist. If the findings fail to reveal added diagnostic value, the claims must contract accordingly. No receipt-free score, invented rank reversal or synthetic human judgment can close these requirements.